Graviton Pressure Theory

The Unified Framework

Individual Submission

This document is part of a multi-part scientific framework

Part 1 of 30

Introduction

to the Unified Framework of the Graviton Pressure Theory

From Description to Mechanism

This submission is part of the broader Graviton Pressure Theory (GPT) project, a redefinition of gravitational interaction rooted in causal field dynamics and physical force transmission. Each document is designed to stand on its own; its full context emerges as part of the larger framework. For the complete series, see https://peopleoftheworld.net/graviton-pressure-theory/.

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1.1 The Return of Causality: Purpose and Position of This Framework

Gravity has always been described. It has rarely been explained.

We can predict, to many decimal places, where a planet will be, how a falling body will accelerate, and how much light will bend as it passes the Sun. What we cannot say—in the dominant theories of physics—is *what acts*. What touches the falling apple? What is exchanged between the Earth and the Moon? What presses a person into a chair?

Graviton Pressure Theory (GPT) is an attempt to answer those questions: a physically grounded model of gravity that restores force, direction and structure to a domain long abstracted into geometry and silence. Where Newton described gravitational attraction without a mechanism, and Einstein replaced attraction with the curvature of spacetime without naming a carrier, GPT proposes a third path: **gravity as the pressure of a physical medium acting on matter**.

GPT does not seek to discard its predecessors. It seeks to complete them.

- Newtonian force is retained—but given a carrier.
- Einsteinian geometry is honoured—but as the mathematics of a physical flow, not a substitute for one.
- Predictive accuracy must be preserved—but no longer at the cost of meaning.

Purpose: This framework exists to replace description presented as cause with mechanism that can be tested.
Scope: It addresses gravitational interaction across scales—from the weight we feel standing still, to tides, orbits, light and time, and into the structure of matter, magnetism and planetary motion.
Objective: To construct a model of gravity that is testable, visualisable, and rooted in a named physical interaction: what touches what, and what is exchanged.

1.2 Description and Mechanism

The central lesson of this framework—learned in writing it—is that physics has, over three centuries, come to accept *description* in place of *mechanism*, and then to forget that it made the exchange. Part 2 traces that history in detail.[1] Here it is enough to state the distinction plainly, because everything that follows depends on it.

To **describe** a phenomenon is to state how it behaves: what follows what, by how much, under which conditions.
To give its **mechanism** is to name what acts: the physical interface between the things involved—what touches what, and what is exchanged.

Both are valuable. A description that predicts is a great achievement. The error is not description. The error is presenting a description as though it were the mechanism, and then treating the question of mechanism as answered, or as naive.

This framework applies one test throughout, to other theories and to its own claims alike:

[1] Rashada, 2026d.

The interface test. Does the explanation name the physical interface—what touches what, and what is exchanged—or does it only describe the action? Where it only describes, the gap is marked as a gap. A name is not a mechanism.

A complete theory of gravity must pass this test for everything gravity is observed to do—not only for orbits and light, but for weight, load, resistance and support, which are how gravity is encountered every moment of every day. Those obligations are set out, item by item, in the companion document *The Gravity Constraint Set*, and General Relativity is audited against them in *The GR Silence Map*.[2] The audit does not claim that General Relativity is wrong. It records where it speaks, where it is silent, and what is customarily substituted where a mechanism would be.

1.3 Historical Arc: From Mystery to Structure

The demand for a mechanism of gravity is not new. It is as old as the law of gravity itself.[3]

- **Newton** gave the law of universal gravitation and openly declined to give its cause. Privately he called action at a distance through empty space "so great an absurdity" that no competent thinker could accept it;[4] publicly, he would frame no hypotheses.[5]
- **Pressure and push models** followed almost at once. Nicolas Fatio de Duillier in 1690, and Georges-Louis Le Sage in the eighteenth century, proposed that gravity is the shadowing of a universal flux of tiny particles—bodies pushed together, not pulled.[6] These models were abandoned in the nineteenth century, chiefly because a moving body should feel drag from such a flux, and because absorbing it should heat matter enormously. Maxwell set out the heating objection himself.[7]
- **Einstein** reimagined gravity as the curvature of spacetime.[8] The theory is extraordinarily accurate. It does not say what, physically, does the curving, or what acts on a body at rest to start it moving. Its own summary is relational: spacetime tells matter how to move; matter tells spacetime how to curve.[9] Einstein himself told an audience in 1920 that in General Relativity space has physical qualities, and that in this sense "space without ether is unthinkable."[10]
- **Modern cosmology** preserves its descriptions by adding dark matter and dark energy, inferred from the gap between prediction and observation. In the standard model they make up about 95% of the universe's energy content; ordinary matter is about 5%.[11] After decades of searching, there is still no confirmed direct detection of a dark matter particle.

GPT stands in the lineage of the push models, and it must answer their objections, not ignore them. Where this framework believes it does, it says so; where the work is not yet done, it says that too.

[2] Rashada, 2026b.
[3] Jammer, 1957.
[4] Newton, 1693.
[5] Newton, 1713.
[6] Edwards, 2002.
[7] Maxwell, 1875.
[8] Einstein, 1916.
[9] Misner et al., 1973; Wheeler, 1990.
[10] Einstein, 1920.
[11] Planck Collaboration, 2020; Peebles and Ratra, 2003.

1.4 What Graviton Pressure Theory Proposes

In outline—each part of the framework develops one layer in detail:

- Gravity is not a pull. It is an **external pressure**: bodies are pressed toward one another by a surrounding medium. **[Hypothesis]**
- The medium is made of **gravitons**—real, directional carriers of momentum. In mainstream physics, the graviton is a hypothetical quantum of the gravitational field, not yet detected;[12] in GPT it is the physical medium itself. **[Hypothesis]**
- Matter interacts with the medium continuously: taking gravitons in, and returning them in a changed, inactive state. Around every mass this lowers the active pressure of the medium, and the higher pressure beyond presses inward. Gravitational behaviour arises from this **differential pressure**, and mass is the measure of how strongly matter takes part in the exchange. **[Hypothesis]**
- Weight, tides, orbits, the bending of light and the slowing of clocks are manifestations of one structured interaction in an ever-present field—not separate phenomena requiring separate explanations. **[Hypothesis]**

These proposals are developed in the parts that follow, and the current working model of the medium—its properties, and how they act together—is set out in *The Mechanics of the Medium*.[13]

The universe, on this view, does not bend. It presses.

1.5 Scientific Integrity and Transparent Method

GPT is not a theory hidden behind inaccessible mathematics. It is a framework built for scrutiny and replication. Every derivation, analogy and prediction is open to challenge. This framework keeps the historical lineage of ideas in view—acknowledging prior models, correcting their assumptions, and preserving the truths they contain.

GPT proceeds by the following commitments:

- No invisible abstraction offered as a mechanism.
- No field effect without a medium or structure that carries it.
- No force without energy; no motion without transfer.
- No stopping point disguised as an answer: where an explanation reaches a primitive, it is marked as the place where a deeper explanation has not yet been found—not as the place where none exists.

How claims are labelled

So that readers can see at a glance what is claimed and on what footing, the second edition of this framework labels its claims about GPT by status:

- **[Established]** accepted fact, independent of GPT;

[12] Zee, 2010.
[13] Rashada, 2026c.

- **[Argued]** a conclusion this framework argues in detail, which readers can check for themselves;
- **[Hypothesis]** a proposal of GPT, not yet tested;
- **[Open]** a question the framework has not yet answered.

Two registers

The physics papers of this framework make no claims about purpose or intent. They stand or fall on mechanism and measurement. The philosophical case—that the removal of cause and mechanism also removed mind, meaning and purpose from our picture of the universe—is made separately, in Part 2, in the register where it belongs. The two are related, but neither is allowed to borrow authority from the other.

This is not yet a confirmed theory. It is a scaffold of causal reinterpretation, rooted in what is observed and built to be tested.

1.6 Where the Work Stands

Honesty about the state of the work is part of the method.

- General Relativity predicts gravitational phenomena with great accuracy and does not name a physical carrier of gravitational action. **[Established]**
- Treating GR's description as the complete mechanism of gravity is a category error, not a finding. This is argued in detail in the papers on the Logical Closure and the Operational Constraints of the Equivalence Principle. **[Argued]**[14]
- GPT accounts qualitatively for the phenomena GR describes, and proposes explanations for the readings attributed to dark matter and dark energy without adding unseen matter. **[Hypothesis]**
- That GPT matches GR's quantitative predictions—the bending of light, the advance of Mercury's perihelion, the timing of clocks—has not yet been shown by independent derivation. Where GPT's numbers currently agree with GR's, some coefficients have been set to match, rather than derived from the medium's properties. **[Open]**
- The historic objections to push models—drag on moving bodies and the heating of matter—are addressed by specific proposals in this framework: that uniform motion through the medium produces no net drag, and that the energy of the exchange is carried away by the returning, inactive medium rather than deposited as heat. The derivations that would confirm these proposals remain to be done. **[Hypothesis]** **[Open]**
- Distinctive predictions, where GPT and GR should differ measurably, are collected in Part 29, *Experimental Divergence*. Until such tests are carried out, GPT remains a hypothesis. **[Established]**

A framework that marks its own gaps invites others to close them. That is the intention.

[14] Rashada, 2026a.

1.7 Why This Framework Matters Now

A theory of gravity that relies on unseen matter and unexplained energy for most of the universe's content may be describing the universe faithfully. Or it may be telling us where its description has run out. Either way, the question of mechanism deserves to be asked again—openly, rigorously, and without the charge of naivety that has kept it closed.

GPT asks it. And it restores three commitments that physics once took for granted:

- **That every motion has a cause.**
- **That no force appears without energy.**
- **That matter and field are not separate, but joined through pressure, exchange and coherence.**

If GPT is right, it is more than another theory of gravity. It is a reunification of structure and mechanism. If it is wrong, the questions it asks will still need answers—and the standard it sets, that description must not pass for explanation, will still apply to whatever replaces it.

1.8 What Follows

The thirty parts of this framework build the case in layers:

- **Foundations** (Parts 2–4): how mechanism was set aside; an inventory of everything gravity actually does; and the path from Newton to GPT.
- **The core theory** (Parts 5–8): gravity as pressure; work, force and stability rebuilt causally; the illusion of pull; and the points where GR and GPT cannot both be right.
- **Evidence and extensions** (Parts 9–14): existing observations read through GPT; applied concepts; four famous classroom demonstrations re-examined (the sealed box, the feather and the ball, the comb and the water, and magnetism); the limits of materialist epistemology; the Moon and gravimetric modulation; and the transition into the detailed theory.
- **The medium in detail** (Parts 15–22): gravitons; the properties of gravitational fields; mass; time; coherence; corridors and lattice resonance; and magnetism as gravimetric resonance.
- **Formalism and tests** (Parts 23–29): the mathematical foundation; units, constants and conversions; planetary mechanics; natural force re-imagined; resonance transmission; and experimental divergence.
- **Conclusion** (Part 30).

Where others wrote geometry, GPT proposes pressure.
Where others drew curves, GPT maps flows.
Where force became an abstraction, GPT seeks to make it real again—and to show its work.

This is gravity, offered for explanation.

Welcome to the pressure field.

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